CS110 · GAME DEVELOPMENT PIPELINE · WEEK 10

# Game & Engine Research

Find out how games work before choosing the tools to build one.

Action 10A · Tuesday, October 27  Due tonight · 11:59 PM Pacific

> **Today: research → compare → discuss → receipt.**
> Keep your notes short. You do not need to install an engine or build a game for Action A.

10A.1 · OBSERVE

## Research one game

▶ **00 / PRE-START · Get ready**

1. Choose **one game** you know or want to investigate.
2. Watch a short gameplay clip or look at its official page.
3. Write the goal, one player action and the feedback the player sees.




**Your game notes**

Game: Half-Life
Goal: Escape the dangers within Black Mesa
Player Action: Utilize weapons to defeat enemies, and use tools to leave different levels of Black Mesa
Feedback: The game was sold well and acclaimed by many critics for its playground-like environment and story.
Source link: https://en.wikipedia.org/wiki/Half-Life_(video_game)

Example: “The goal is to reach the exit. I press jump. The character rises and a sound plays.” A few clear lines are enough.

10A.2 · CHOOSE THE TOOL

# Compare three engines

A **game engine** supplies systems such as graphics, sound, input and physics. The **editor** is where you arrange and change your project.

## Godot

2D and 3D. Uses nodes and scenes. C# requires the .NET edition.

Nodes & scenes ↗
C# support ↗

## Unity

2D and 3D. Uses GameObjects and components. Supports C# scripts.

GameObjects ↗
Scripting ↗

## Stride

A C# game engine. Uses scenes, entities and components.

Game Studio ↗
Official site ↗

Use these three, or choose another engine discussed in class. For each, record **2D/3D, its programming language, one useful feature and an official source link**. Then choose one you would try and give one reason.

**Your three-engine comparison**

https://en.wikipedia.org/wiki/Godot_(game_engine)
2. Unity/3D & 2D/C#/Generative AI Tools/ https://en.wikipedia.org/wiki/Unity_(game_engine)
3. Stride/3D/C#/.NET foundation/ https://en.wikipedia.org/wiki/Stride_(game_engine)

I will be using Unity, because it is the most popular game engine, and the engine use C# coding.

▶ **What if I cannot find which engine made my game?**

10A.3 · EXPLAIN IT TOGETHER

# Discuss one game object

With a partner or small group, draw **input** → **action** → **feedback** on the whiteboard or shared page. Use an object from your selected game.

## Data: what it knows

A player object might store **health = 100** and **speed = 5**.

## Action: what it does

A method such as **Jump()** describes an action. A class describes a kind of object; each instance keeps its own state.

1. Name one object, two values it stores and one action it performs.
2. Explain your engine choice to a classmate.
3. Save a board photo or an exported shared diagram. Write one useful comment from your partner.

**Your discussion notes**

Object: Headcrab
Two Data Values: (Health = 25) (Speed = 10)
Action: Kill the headcrab with crowbar
My engine choice and Reason: We chose Unity because the engine uses C# language that is suited perfectly.

An accessible text diagram can replace a board photo. If you are working alone, use the instructor's check-in. Ask questions during the lecture; this is guided practice.

▶ **AI and research help**

10A.4 · BRING BACK THE RECORD

# Download, push, submit

Your notes become your README. You do not need to write a second report.

**Your name**

Samuel Leal

**Your CS110 GitHub repository URL**

https://github.com/Lexicon-Lexxie/cs110-week10.git

- [x] I researched one game and recorded a source.
- [x] I compared three engines and explained my choice.
- [x] I saved discussion/board evidence and an object example.
- [x] I checked my notes and am ready to save my receipt.

1. Save **README.md**, your **board/diagram evidence** and your **10A receipt** in `Week10/Action10A/` .
2. **Commit and push** to your own CS110 GitHub repository. GitHub's Upload files also works.

3. Open your repository online and confirm the files are present.
4.

**Due today: October 27, 11:59 PM Pacific.** Each learner submits their own repository URL, even when discussing as a group.

The receipt records your notes and completion declarations. It does not verify attendance, correctness, Canvas submission or a grade. Download your files before leaving a school computer; browser notes may be erased.

▶ **Next: Action B · Wednesday, October 28**